Tensor computations

Infinite mean condition number in every identifiable tensor format

Difficulty: challenging **Importance:** interesting to the community

Status: open conjecture; explicit journal source published in 2023.

Context and notation

Fix $d \geq 3$, $n_j \geq 2$, and $r \geq 2$. Assume generic complex identifiability: outside a proper algebraic exceptional set, a complex tensor of rank r in this format has a unique unordered collection of r rank-one summands. Let M_r be the smooth identifiable locus of real rank-r tensors, with induced Euclidean volume dV. The random input has density

$$d\mu(A) = Z^{-1}e^{-\|A\|_F^2/2}\,dV(A).$$

For the addition map $\Phi(a_1,\dots,a_r) = \sum_i a_i$ on rank-one tensors, let Ψ be a local inverse at A. Use Frobenius norms and their product norm. Define

$$\kappa(A) = \|D\Psi(A)\|_2, \qquad \kappa_{\mathrm{ang}}(A) = \|D(p^{\times r} \circ \Psi)(A)\|_2, \quad p(a) = a/\|a\|_F.$$

Values on measure-zero exceptional sets do not affect the expectations. This samples tensors by volume, not independent Gaussian summands.

Problem statement

Under this probability model, prove or disprove

$$\mathbb{E}_\mu \kappa(A) = \infty$$

for every admissible format and $r \geq 2$. Rank two is already proved. The unresolved task removes the additional smaller-format identifiability assumption used for the higher-rank theorem; it must not be counted again for parameter ranges covered by that theorem.

Reference

Carlos Beltrán, Paul Breiding, and Nick Vannieuwenhoven, *The Average Condition Number of Most Tensor Rank Decomposition Problems Is Infinite*, *Foundations of Computational Mathematics* 23 (2023), 433–491: Definition 1, Assumption 1, equation (5), Theorems 1–2, and Conjecture 1. The author preprint labels the conjecture 1.8.

Status check

Searches included *"average condition number" "tensor" conjecture 2025 2026* and *"regular condition number" "tensor" "2026" conjecture proof*. No removal of the remaining hypothesis was located. The latest explicit conjecture located is the 2023 journal version.